

Draft - expert review pending

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Unit 5 assessment

Unit summary

Data are raw facts collected for a question, by observation, survey or records; tallies and frequency tables organize them so summary is simple counting, and a table whose total does not close names a collection problem. Each display answers one kind of question - bar charts compare categories, pie charts show shares of one whole, line graphs track change over time - and reading a display means reading its construction. The mean is the fair share outliers drag, the median the middle that ignores them, the mode the most frequent value; the honest average depends on the data. A published statistic is read with five questions: who collected it, how, compared to what, on what base, and what is missing - and Sindh's 57.54 percent literacy against a 46.29 percent out-of-school rate shows how much one number can hide.

Summative assessment

Multiple-choice questions (MCQs)

1. The five stages of the data cycle, in order, are: (*Remember*) a) collect, question, summarize, organize, interpret b) question, collect, organize, summarize, interpret c) question, organize, collect, interpret, summarize d) collect, organize, question, summarize, interpret
2. To learn how many pupils arrive after the bell each day this week, the best method is: (*Apply*) a) a survey of pupils b) observation at the gate c) the census report d) a pie chart
3. A survey of 40 returns frequencies 12, 14 and 9. The table tells you: (*Apply*) a) the survey is complete b) 5 responses are unrecorded c) the mode is 14

d) the mean is 35

4. The display that best shows each travel mode's share of the class is: *(Apply)*
a) a bar chart b) a line graph c) a pie chart d) a tally
5. The display that best tracks attendance across a week is: *(Apply)* a) a pie chart b) a line graph c) a pictograph d) a frequency table
6. A category that is 30 percent of the whole takes a pie-slice angle of: *(Apply)*
a) 30 degrees b) 90 degrees c) 108 degrees d) 144 degrees
7. A bar chart whose vertical axis starts at 10 instead of 0: *(Understand)* a) is correct if labelled b) exaggerates small differences between bars c) makes bars harder to read d) shows shares of a whole
8. For the takings 4,000, 4,200, 4,100, 4,300, 39,000 rupees, the honest "typical day" is the: *(Apply)* a) mean, 11,120 rupees b) median, 4,200 rupees c) mode, 4,000 rupees d) total, 55,600 rupees
9. The only average that works for categories (like travel modes) is the: *(Remember)* a) mean b) median c) mode d) range
10. Sindh's 2023 literacy rate of 57.54 percent is measured on the base: *(Understand)* a) all ages b) 10 years and above c) 15 years and above d) enrolled pupils only

Restricted-response questions (RRQs)

1. Name the three data collection methods and one school question each suits. *(Understand)*
2. A survey of 40 gives frequencies 12, 14, 9. State what the total tells you and the next move. *(Analyze)*
3. Compute the mean, median and mode of: 20, 45, 50, 55, 60, 62, 62, 70, 70, 70, 96. *(Apply)*
4. A pictograph shows 3 bus icons for Class A and 5 for Class B, key "one icon = 10 pupils". State the counts and one way it misleads without the key. *(Analyze)*
5. For the class strengths 38, 42, 35, 40, 45: compute the median and the mode, and state which classes (if any) share it. *(Apply)*

6. A bar chart's axis starts at 10. State what the picture exaggerates, in one sentence. *(Understand)*
7. Compute the pie-slice angle for a category that is 22.5 percent of the whole, showing the working. *(Apply)*
8. Explain in two sentences why the median is preferred to the mean for house prices or incomes. *(Understand)*
9. Sindh's male literacy is 64.23 percent and female 50.21 percent. Compute the gap, and say what the single headline number 57.54 hides. *(Analyze)*
10. Name the five questions to ask of any published statistic. *(Remember)*

Extended-response questions (ERQs)

1. The headteacher wants to know the school's most common travel mode. Design and run a mini-survey of 20 pupils (fixed choices: walk, van, rickshaw, bicycle, other): present your frequency table with the total check, state the most common mode and its percentage, and identify one way your collection could be biased and how you would fix it in a rerun. *(Analyze)*
2. The school's five class strengths are 38, 42, 35, 40 and 45. Name the best display for "which class is largest and by how much" and draw or describe it precisely; name one display that would be a poor choice and say what it would hide or exaggerate; and compute the pie-slice angle for the largest class's share of total pupils, showing the working. *(Analyze)*
3. The 2023 census reports Sindh's literacy (10 years and above) as 57.54 percent overall, 64.23 percent male and 50.21 percent female, against 54.57 percent in 2017; the out-of-school rate (ages 5 to 16) is 46.29 percent (PBS, 2023). Compute the male-female gap in percentage points; state the rise since 2017 and say whether it is large or small compared with the gap; and write the two interpretation questions a teacher should ask before quoting the 57.54 percent figure to a parents' meeting, and what each guards against. *(Analyze)*
4. Miss Farhana's class of 11 scored 20, 45, 50, 55, 60, 62, 62, 70, 70, 70, 96. Compute all three averages; explain what the ordering of mean below median below mode reveals about the class; and write the two sentences you would say at a parents' meeting about which average you are quoting and why. *(Analyze)*
5. Using the frequency table your group built in the Topic 5.1 activity (or a fresh survey of 20 classmates), present the whole data cycle for your own

question: the question, the collection method and its bias risk, the organized table, the best display for it, the honest average (if the data are numerical), and your interpretation - then evaluate which stage of your cycle was weakest and what you would redo. (*Evaluate*)

Answers and marking guidance

MCQ answer key

1. **b** - question, collect, organize, summarize, interpret.
2. **b** - watching arrivals records what actually happens; a survey asks what pupils remember.
3. **b** - $12 + 14 + 9 = 35$, so 5 of 40 are unrecorded.
4. **c** - shares of one whole are the pie chart's job.
5. **b** - change over time is the line graph's job.
6. **c** - $30\% \times 360 = 108$ degrees.
7. **b** - the truncated axis stretches small differences into large-looking ones.
8. **b** - one wedding-order outlier drags the mean to 11,120, which no ordinary day resembled.
9. **c** - the mode is the most frequent value and works for categories too.
10. **b** - the census literacy rate counts persons 10 years and above.

RRQ model answers and mark schemes

1. Model: observation (arrivals at the gate), survey (which snack to stock), records (last month's absences) - or equivalent pairings with reasons.
Marks (3): 1 per method with a fitting question.
2. Model: the total 35 against the sample 40 shows 5 responses unrecorded; the next move is to find them (or rerun), not to summarize. **Marks (2):** 1 - the gap identified; 1 - the correct next move.
3. Model: total 660, mean 60; median 62 (sixth of 11 sorted); mode 70 (three times). **Marks (3):** 1 each.
4. Model: Class A 30, Class B 50; without the key the icons suggest a 3-to-5 count (a 20-pupil error). **Marks (2):** 1 - both counts; 1 - the missing-key danger.

5. Model: sorted 35, 38, 40, 42, 45: median 40; no repeated value, so no mode. **Marks (2):** 1 - median 40; 1 - no mode stated.
6. Model: differences between bars look larger than the values justify, because the bars' visible heights start from 10. **Marks (2):** 1 - exaggeration named; 1 - mechanism.
7. Model: $22.5\% \times 360 = 81$ degrees. **Marks (2):** 1 - method; 1 - value.
8. Model: a few very high prices drag the mean above what almost any house costs; the median reports the middle of the market, unaffected by the extremes. **Marks (2):** 1 - outlier effect on mean; 1 - median's resistance.
9. Model: $\text{gap} = 64.23 - 50.21 = 14.02$ percentage points; the headline 57.54 hides the near-10-point-plus divide between male and female literacy (and the 46.29 percent out-of-school rate). **Marks (2):** 1 - gap computed; 1 - what is hidden named.
10. Model: who collected it, how was it collected, compared to what, what is the base, what is missing. **Marks (5):** 1 per question.

ERQ rubrics

1. Travel survey. Analytic rubric: *Table with check* (0-2: correct table totalling 20); *Mode and percentage* (0-2: most common mode with a correct percentage); *Bias identified* (0-2, **Analyze:** a real collection bias with a concrete fix). Total 6.
2. Class strengths. Analytic rubric: *Best display* (0-2: bar chart, precisely described, axis from zero); *Poor choice analysed* (0-2, **Analyze:** e.g. a pie chart hides "by how much"); *Slice angle* (0-2: total 200, $45/200 = 22.5\%$, angle 81 degrees with working). Total 6.
3. Sindh literacy. Analytic rubric: *Gap* (0-2: 14.02 percentage points); *Rise compared* (0-2, **Analyze:** 2.97 points, dwarfed by the gap at more than four times its size); *Questions* (0-2: two interpretation questions with correct guards). Total 6.
4. Marksheet. Analytic rubric: *Averages* (0-3: mean 60, median 62, mode 70, 1 each); *Ordering explained* (0-2, **Analyze:** mean below median below mode reveals a low tail pulling the mean down); *Parents' sentences* (0-2: names the average quoted with an honest reason). Total 7.
5. Own data cycle. Analytic rubric: *Cycle presented* (0-3: question, method, table, display all present and fitting); *Honest average* (0-2: correct choice for

the data type); *Interpretation* (0-2: the answer stated with its limits);
Weakest stage evaluated (0-3, **Analyze**: the weakest stage named with
evidence and a redo plan). Total 10.

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